

CorrTrack — candidate selection: v1 degenerates into brute force, v2 does not

v1 L2 radius cell_size — only filter

share of pairs passing each stage · dense data

Sketch



Selection



no intermediate filter · freq_threshold=0 (full_vector)

Pearson validation



$$\|a-b\|^2 \approx 2(1-r) \Rightarrow r \gtrsim 1 - \tau^2/2$$

cell_size	τ	keeps $r \geq$
1	0.78	0.70 ✓
2	1.55	-0.2 (almost all)
3	2.32	all ($r \geq -1$)

Effect — candidates $\approx$ all pairs $\Rightarrow$ exhaustive validation = **brute force in sketch space** (slower than BF). The sweep scans $\text{cell_size} \in [1, 2, 3]$.

normalized RP

$$\|a-b\| \leq \tau$$

exact, $O(n^2)$

v2 bounded cosine gate before Pearson

share of pairs passing each stage · dense data

Sketch



sketch_method (5 variants) — drives separability (AUC)

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Cosine gate



bounded $[-1, 1]$ · always applied (validate_sketches)

Pearson validation



$\cos(a, b) \approx r$ directly — bounded threshold $[-1, 1]$.

- ✓ Gate sketch_threshold cuts **before** the costly Pearson
- ✓ Loose index $\rightarrow$ inflates the cosine (cheap), **not** Pearson
- ✓ Rest = data separability $\rightarrow$ diagnose_select

Effect — set reduced before validation $\Rightarrow$ **Pearson protected.**

RP / fft / median

query_radius (cdist)

$\cos \geq 0.7$

$r \geq 0.8$

KEY POINT

The **v1 mechanism** (unbounded radius, no filter before Pearson) is **absent in v2**. The “almost everything passes” symptom may return in **v2**, but **bounded by the cosine gate** (Pearson protected) and **driven by data separability**.